Sai Balkawade

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Summary

Computer Engineering student focused on full-stack web development, working across React front ends and Node.js/Express back ends with Supabase and PostgreSQL for data and auth. Background includes a frontend internship, 19 merged open-source pull requests, and a few deployed side projects.

Technical Skills

Languages	JavaScript, TypeScript, Python, SQL
Frontend	React.js, Next.js, Vite, Tailwind CSS, State Management
Backend & APIs	Node.js, Express.js, Hono, RESTful APIs, WebSockets, Supabase
Databases & Cloud	PostgreSQL, MongoDB, MySQL, Redis, Cloudflare Workers, Docker, Git, Linux
AI & Tooling	Generative AI, RAG Pipelines, Vector Search (ChromaDB), Postman

Work Experience

Frontend Intern, XIRCLS *Dec 2025 – Mar 2026*
Mumbai, India
- Engineered responsive, reusable UI components for an enterprise CRM platform, translating business specs into production code.
- Integrated frontend modules with backend REST APIs and CRM workflows, improving state handling and UI consistency.

Technical Member, Project Cell, Fr. Conceicao Rodrigues College of Engineering *Aug 2025 – Present*
Mumbai, India
- Architected and deployed the official Project Cell portal on Next.js with modular, performance-tuned component hierarchies.
- Delivered a real-time event management app (React Native, Expo, Firebase) with role-based auth for 120+ active participants.
- Built automated testing and deployment workflows using Playwright and Node.js that streamlined recurring event operations.

Projects

Clip | Zero-Knowledge Pastebin & Real-Time Collab Pad (React, TypeScript, Cloudflare Workers, Hono) Live Demo | GitHub
- **Architecture**: Engineered an ephemeral file and text sharing platform featuring a real-time collaborative Live Pad synchronized over WebSockets.
- **Security**: Implemented client-side zero-knowledge encryption using PBKDF2 and AES-256-GCM; plaintext and keys never touch the server.
- **Edge Infrastructure**: Deployed high-performance backend handlers on Cloudflare Workers with Hono, custom TTLs, ZIP archiving, and CLI/curl support.

LegalAI | AI Legal Research Engine (React, Node.js, ChromaDB, Llama 3.1) Live Demo | GitHub
- **Product**: Developed an AI research platform covering IPC and BNS statutes that delivers citation-backed legal analysis naming exact sections.
- **RAG Pipeline**: Built semantic retrieval across 850+ statutory sections using ChromaDB vector embeddings and Sentence Transformers.
- **API Gateway**: Orchestrated Llama 3.1 70B/8B models through a Node.js gateway with tiered rate-limiting and route optimization.

ReCode | AI DSA Companion Platform (React, TypeScript, Node.js, MongoDB, Redis) Live Demo | GitHub
- **Platform**: Created a full-stack DSA revision web application with automated spaced-repetition schedules to resurface weak topics.
- **AI Integration**: Designed a tiered hint generator (hint, approach, full walkthrough) allowing progressive problem-solving assistance.
- **Performance & Auth**: Cached frequent query lookups in Redis to slash latency; implemented secure authentication using JWT and OTP verification.

Education

B.Tech in Computer Engineering, Fr. Conceicao Rodrigues College of Engineering *2024 – Present*
SGPA: 8.59 / 10 | *Mumbai, India*
Class XII (Computer Science), RCF Junior College, Alibag | 79.67% *2022 – 2024*
Class X, RCF Secondary and Higher Secondary School, Alibag | 93.60% *2021 – 2022*

Achievements

- **Open Source**: 19 merged PRs to Sugar Labs / Music Blocks: fixed memory leaks, refactored `eval()`, and built plugin hooks.
- **Hackathons**: Runner-Up at Web Fusion 2.0 (Thakur Shyamnarayan COE) and 3rd Place at Echelon 2.0 (NMIMS Navi Mumbai).
- **Chess**: 6th Rank in All India Inter-University Championship; West Zone Medallist.
- **Problem Solving**: Solved 200+ DSA problems across LeetCode and competitive programming platforms.